

JOB MARKET ANALYSIS REPORT

Introduction

The objective of this project is to analyse the Data Science job market and identify important trends related to job opportunities, salaries, industries, companies, job roles, education, and technical skills.

The dataset contains **742 job records and 42 features**, including Job Title, Salary Estimate, Company Name, Location, Industry, Rating, salary-related fields, job title categories, seniority, degree information, and multiple skill columns. The purpose of the project is to identify job-market trends, in-demand skills, qualifications, and responsibilities that can help job seekers and employers make better decisions.

The analysis was performed using **Microsoft Power BI**. Power Query was used for data cleaning and transformation, DAX was used for calculations, and three interactive dashboards were created to analyse different aspects of the job market.

Data Preparation and Transformation

Before performing the analysis, the dataset was cleaned using **Power Query Editor**. The data preparation process included checking data types, cleaning text fields, validating duplicate records, and handling missing or unavailable values.

According to the dataset documentation, the value **-1 represents unavailable information**. These values were handled appropriately to avoid incorrect analysis and calculations.

The dataset also contains salary fields including Lower Salary, Upper Salary, and Average Salary. The average salary is calculated using the lower and upper salary values and is expressed in thousands of US dollars.

For skill analysis, the multiple skill columns were transformed using the **Unpivot Columns** feature. The original dataset contains skill indicators such as Python, Spark, AWS, Excel, SQL, Tableau, Hadoop, and other technologies, where **1 indicates that the skill is required and 0 indicates that it is not required**.

A separate **Job Skills table** was created to make it easier to analyse the skills required for different job titles.

DAX measures were created for important calculations such as Total Jobs, Average Salary, Average Minimum Salary, Average Maximum Salary, Total Companies, and Skill Job Count.

Dashboard 1: Job Market Overview

The first dashboard analyses job opportunities across different states, industries, companies, and job titles.

The analysis shows that job opportunities are concentrated in specific states, indicating that certain locations have a higher demand for Data Science and analytics professionals. This suggests that job seekers may find more opportunities by targeting locations with higher job-posting activity.

The industry analysis also shows that Data Science-related roles are not limited to technology companies. Industries such as healthcare, pharmaceuticals, insurance, and technology contribute to the demand for analytical professionals.

The analysis of companies identifies organizations with higher numbers of job openings. Similarly, job title analysis identifies the roles that have greater demand in the market. The dataset includes normalized job-title categories through the `job_title_sim` field.

Key Insights

- Job opportunities are concentrated in certain geographical locations.
- Multiple industries require Data Science and analytical professionals.
- Certain companies show higher hiring activity.
- Some job titles have significantly higher demand than others.

Recommendations

Job seekers should adopt a targeted job-search strategy by focusing on high-demand locations, industries, companies, and job roles. Instead of applying randomly, candidates can use market trends to identify better opportunities.

Employers can also expand their talent search to additional locations and consider remote hiring to access a wider talent pool.

Dashboard 2: Salary and Education Analysis

The second dashboard analyses salary differences across states, job titles, and education categories.

The comparison of average minimum and maximum salaries shows that compensation varies across locations. These differences may be influenced by factors such as industry presence, demand for skilled professionals, and regional market conditions.

The analysis also compares salaries across popular job titles. An important finding is that the job title with the highest number of openings may not necessarily have the highest average salary. Therefore, job availability and salary potential should be considered together.

The dataset also contains information related to educational qualifications, including Master's (M) and Ph.D. (P) categories.

Key Insights

- Salary levels vary across different states.
- Job roles with high demand are not always the highest-paying roles.
- Salary patterns differ according to job title and seniority.

- Education categories can show differences in average salary.

Recommendations

Job seekers should evaluate opportunities using multiple factors rather than focusing only on salary.

A better decision-making approach is:

Job Demand + Salary + Required Skills + Location + Career Growth

Candidates should also consider additional education based on their target career path rather than assuming that an advanced degree alone guarantees higher compensation.

5. Dashboard 3: Skills and Job Intelligence

The third dashboard focuses on identifying the skills required by companies and comparing skill requirements across different job titles.

The skill columns were transformed into a separate Job Skills table using the Unpivot feature in Power Query. This made it possible to analyse the frequency of each skill and identify which skills are associated with specific job roles.

The analysis shows that employers look for a combination of technical and analytical skills. The skill requirements can differ depending on the job role.

For example, some roles may require stronger programming and machine learning skills, while others may focus more on databases, cloud technologies, data processing, visualization, or business intelligence.

Key Insights

- Technical skills play an important role in job-market demand.
- Employers often require a combination of skills rather than a single technology.
- Different job titles require different skill combinations.
- Skill analysis can help identify potential gaps between market requirements and candidate skills.

Recommendations

Candidates should avoid learning tools randomly. Instead, they should first identify their target job role and then develop the skills most relevant to that role.

A recommended approach is:

Target Job Role → Required Skills → Skill Development → Projects → Job Applications

This allows candidates to build a focused learning path and improve their chances of meeting market requirements.

6. Overall Findings and Conclusion

The analysis shows that the Data Science job market is influenced by several interconnected factors, including location, industry, company hiring activity, job role, salary, education, seniority, and technical skills.

The major findings of the analysis are:

- Job opportunities are concentrated in specific locations.
- Data-related roles are available across multiple industries.
- Certain companies and job titles have higher hiring demand.
- Salary varies across states and job roles.
- High job demand does not always mean higher salary.
- Different job roles require different technical skill combinations.
- A role-specific skill development strategy is more effective than learning tools without a clear career objective.

Final Recommendations

Job seekers should use a **data-driven approach** when planning their careers. Instead of focusing on only one factor, they should consider:

Location + Job Demand + Salary + Required Skills + Career Growth